



TRANSPORT LAYER

Title: Elements of Transport Protocols
Congestion Control

Subject: Fundamentals of Computer Networks

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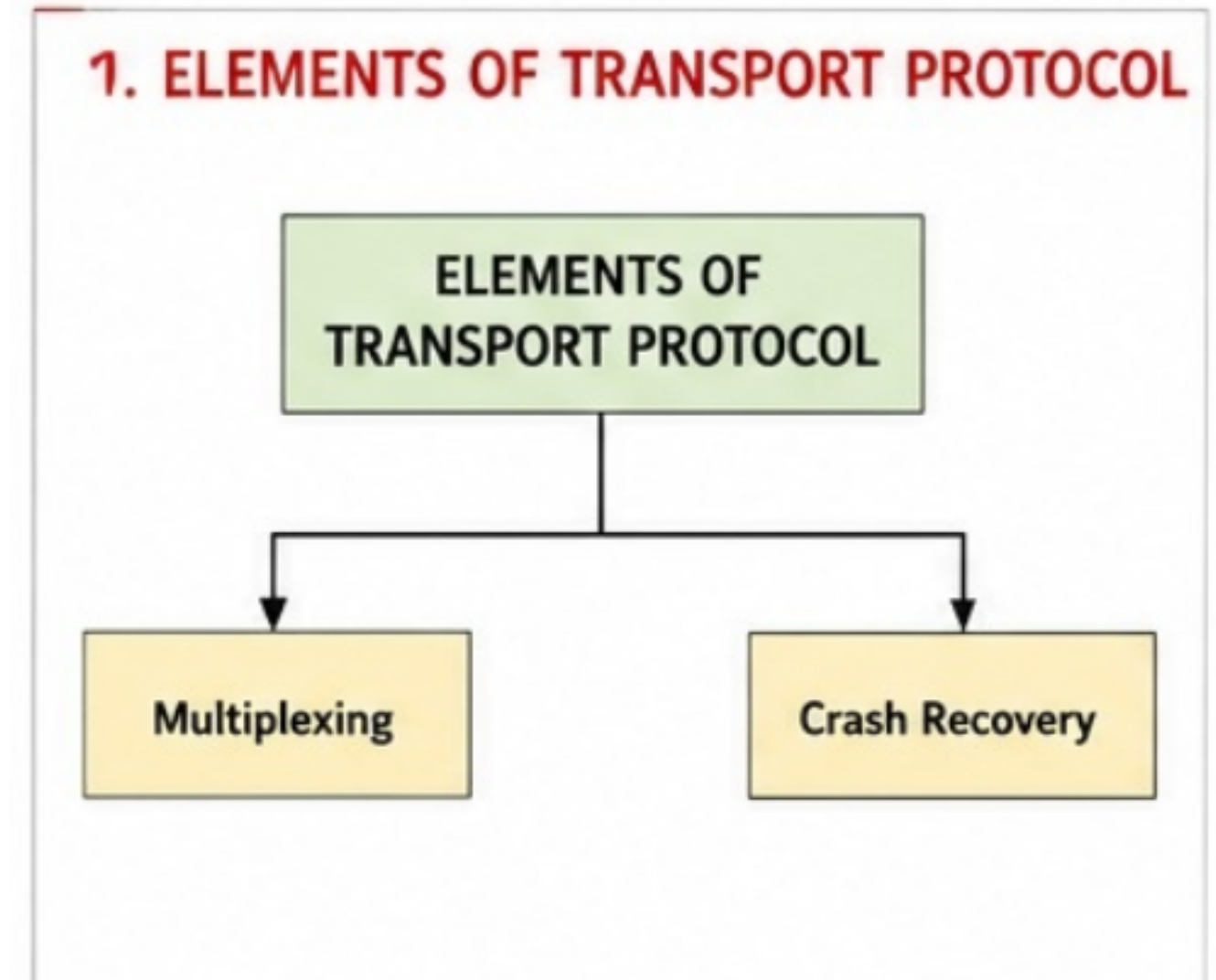
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Elements of Transport Protocol

- Transport protocol provides reliable end-to-end communication
- Manages communication between sender and receiver
- Ensures proper delivery of transport layer segments
- Handles connection management and recovery
- Improves communication efficiency and reliability
- Contains several important operational mechanisms

Main Elements

- Multiplexing
- Crash Recovery

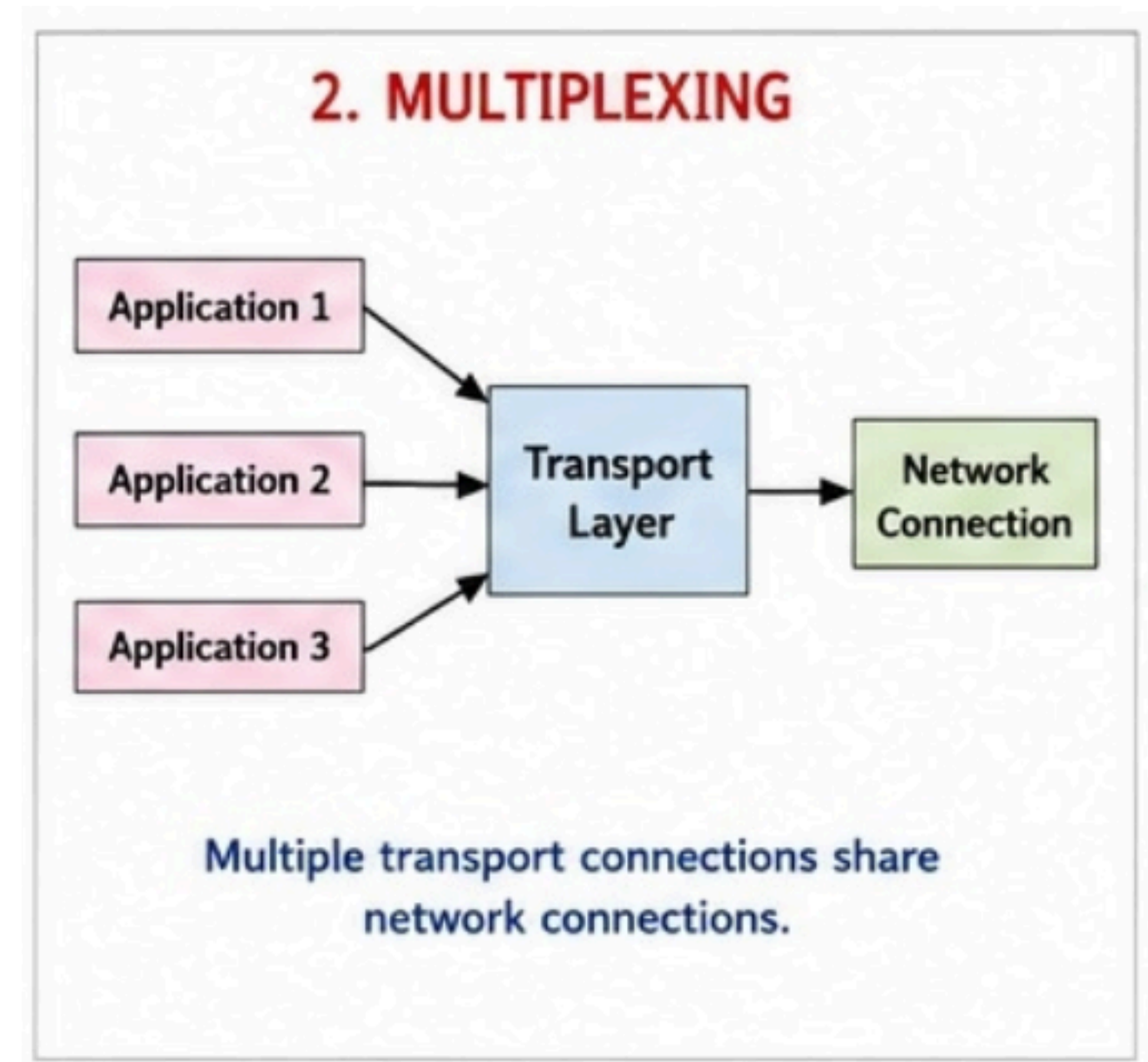


Multiplexing

- Multiple transport connections share network connections
- Improves bandwidth utilization
- Reduces communication cost
- Allows efficient resource sharing
- Supports simultaneous communication
- Helps improve network performance
- Used to increase communication efficiency

Types of Multiplexing

- Upward Multiplexing
- Downward Multiplexing

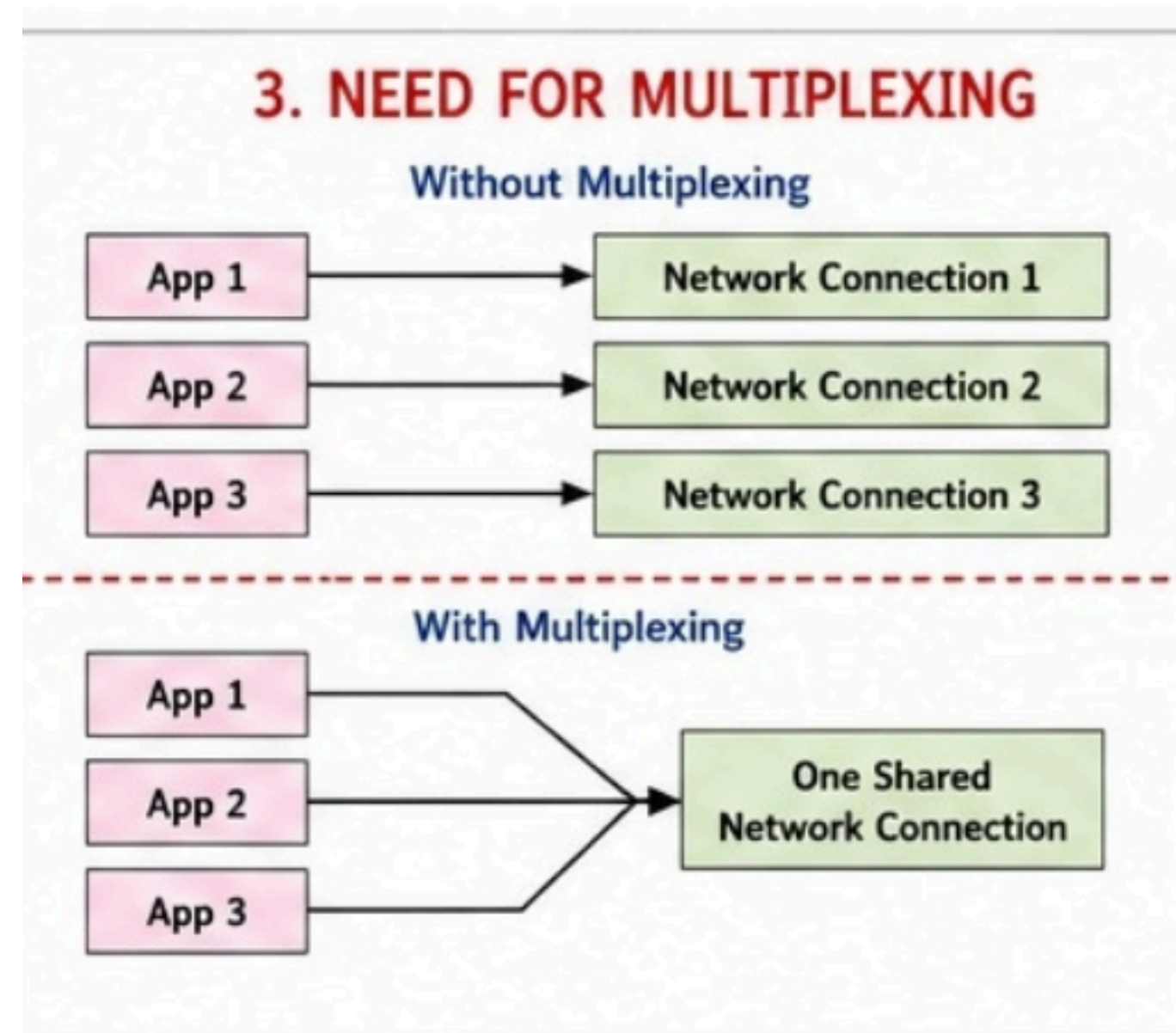


Need For Multiplexing

- Reduces number of physical network connections
- Enables sharing of available bandwidth
- Improves communication efficiency
- Minimizes resource wastage
- Reduces transmission overhead
- Supports multiple users simultaneously
- Makes network utilization more efficient

Advantages

- Better bandwidth utilization
- Cost reduction
- Efficient communication
- Resource optimization

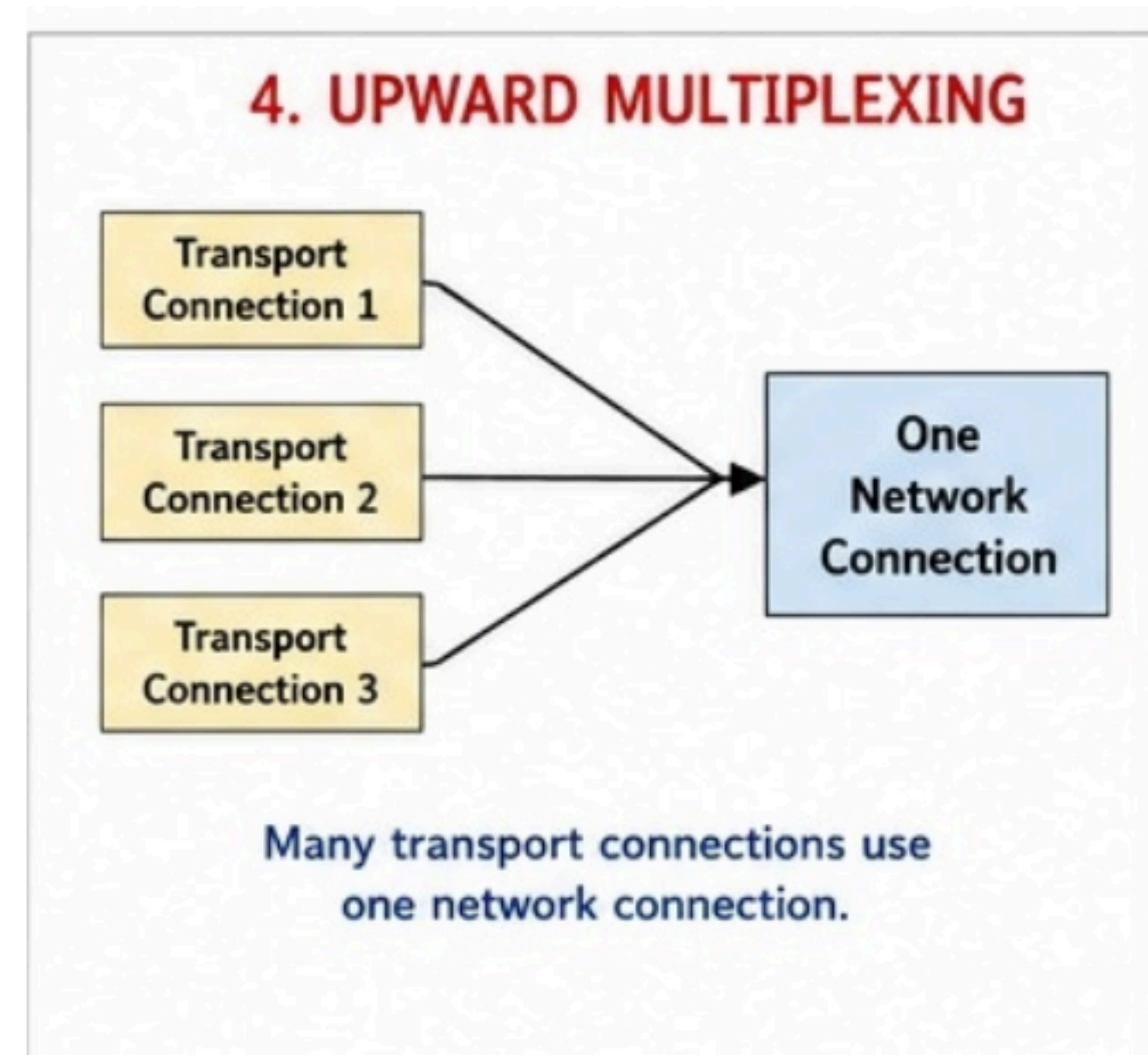


Upward Multiplexing

- Multiple transport connections use one network connection
- Saves bandwidth and network resources
- Reduces communication overhead
- Efficient for low traffic communication
- Improves utilization of network connection
- Commonly used in transport protocols
- Many users share single communication channel

Important Concept

- Many-to-One Communication

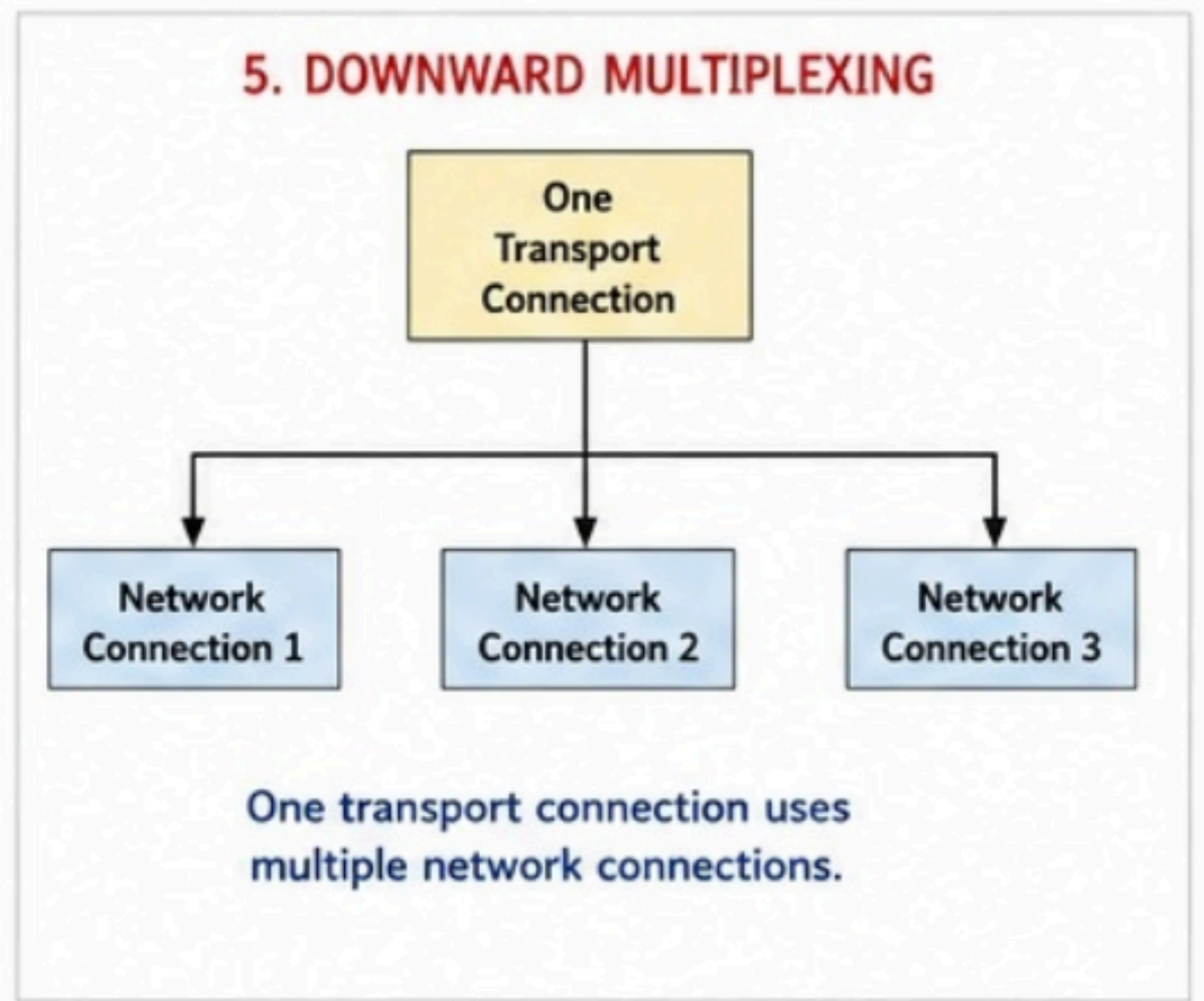


Downward Multiplexing

- One transport connection uses multiple network connections
- Improves communication reliability
- Increases throughput and performance
- Supports parallel data transfer
- Provides better fault tolerance
- Multiple network paths improve efficiency
- Useful for high-speed communication

Important Concept

- One-to-Many Communication



Comparison of Multiplexing Types

Upward Multiplexing	Downward Multiplexing
Multiple transport connections	Single transport connection
One network connection	Multiple network connections
Saves bandwidth	Improves reliability
Reduces overhead	Increases throughput
Many-to-one communication	One-to-many communication
Better resource utilization	Better performance

6. COMPARISON OF MULTIPLEXING TYPES		
Feature	Upward Multiplexing	Downward Multiplexing
Transport Connections	Many	One
Network Connections	One	Many
Communication	Many-to-One	One-to-Many
Resource Usage	Saves Resources	Uses More Resources
Reliability	Less	More
Throughput	Less	More
Overhead	Less	More
Use	Low Traffic	High Traffic

Important Point

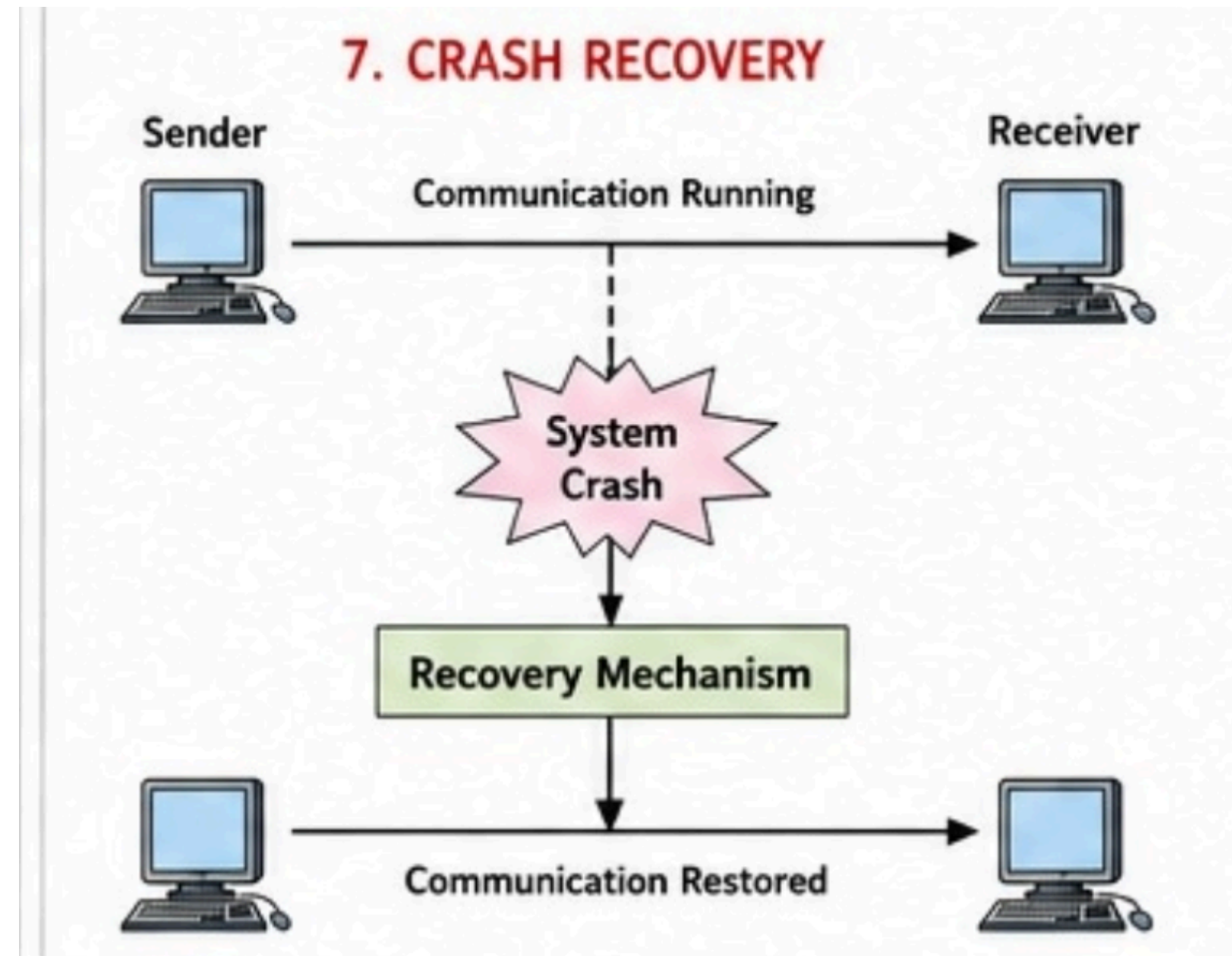
- Different multiplexing methods improve network efficiency differently

CRASH RECOVERY

- Handles unexpected system failures
- Restores interrupted communication
- Prevents duplicate data transfer
- Maintains communication consistency
- Detects incomplete connections
- Ensures reliable communication after restart
- Helps recover safely after failures

Causes of Crash

- Power failure
- System failure
- Network interruption

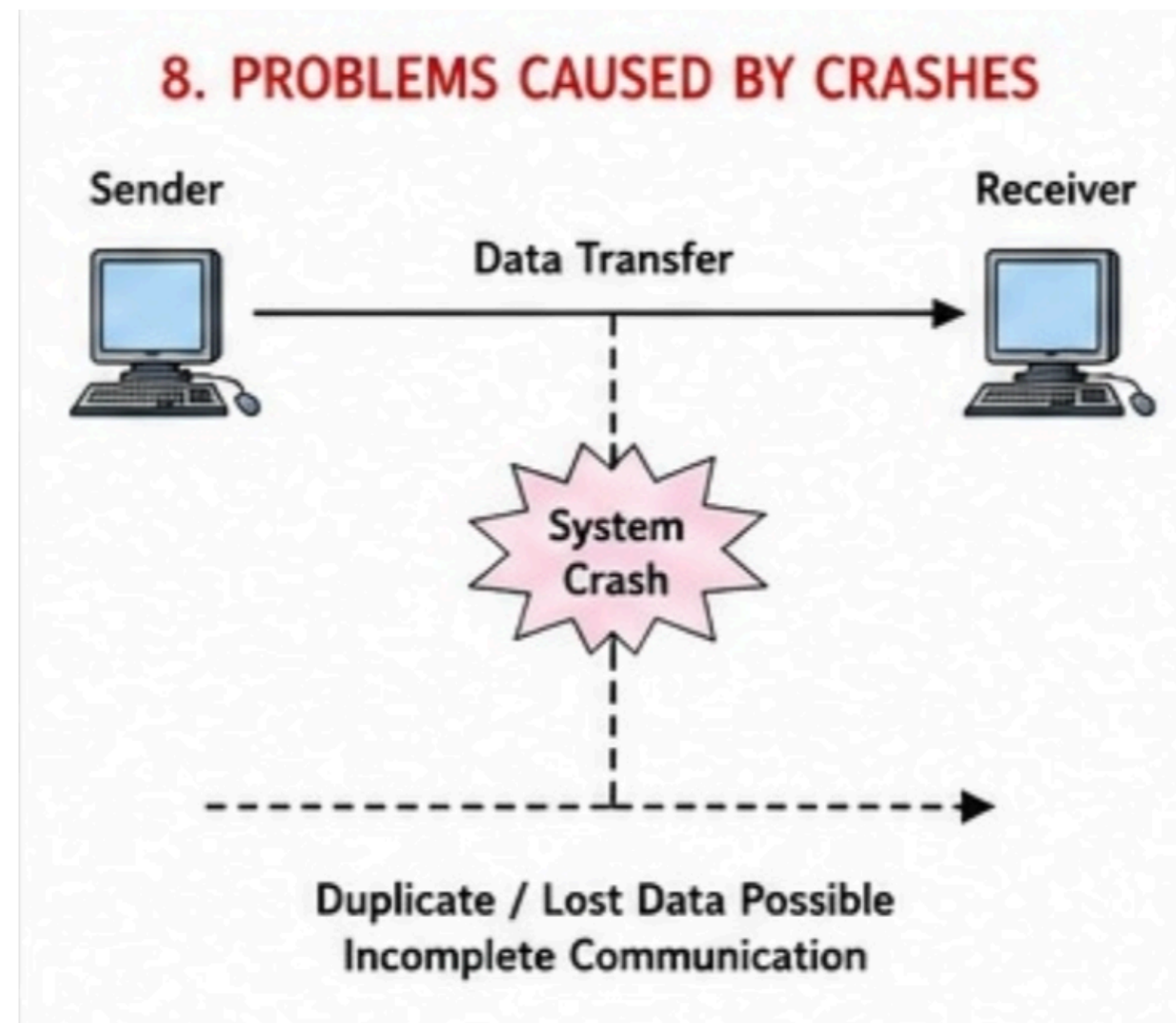


PROBLEMS CAUSED BY CRASHES

- Connection state information may be lost
- Duplicate packets may occur
- Incomplete communication may happen
- Data retransmission problems may arise
- Receiver may get duplicate segments
- Synchronization between hosts may fail
- Communication consistency may be affected

Challenges

- Detecting interrupted communication
- Recovering data safely
- Avoiding duplicate transmission

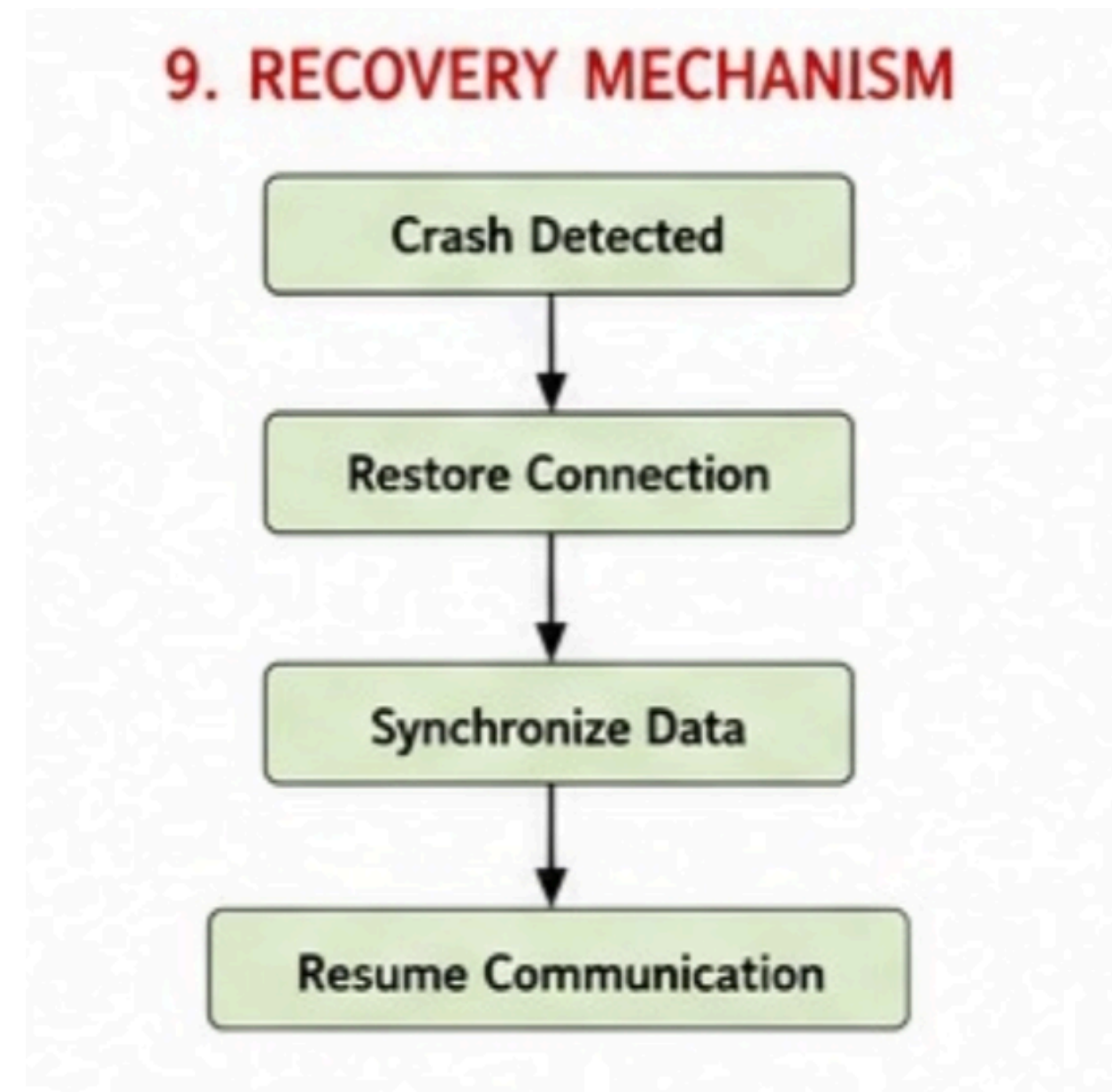


RECOVERY MECHANISM

- Detects interrupted communication
- Restores communication safely
- Synchronizes sender and receiver
- Avoids duplicate data delivery
- Resumes interrupted communication
- Maintains reliable data transfer
- Ensures stable communication after restart

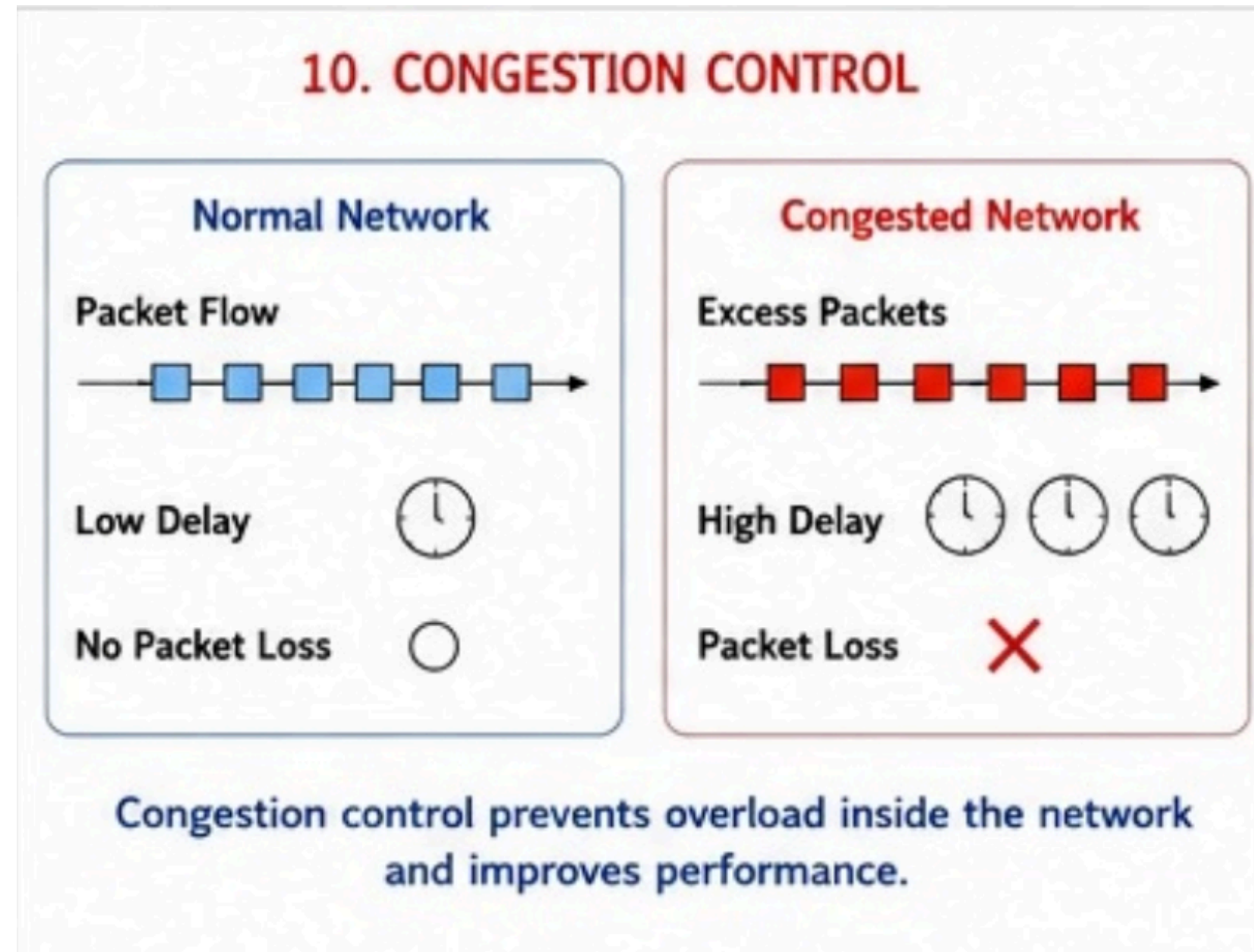
Recovery Steps

- Detect crash
- Restore connection
- Synchronize communication
- Resume transmission



CONGESTION CONTROL

- Prevents overload inside the network
- Controls traffic entering the network
- Reduces packet loss
- Minimizes transmission delay
- Improves overall network performance
- Maintains efficient communication
- Prevents network congestion collapse

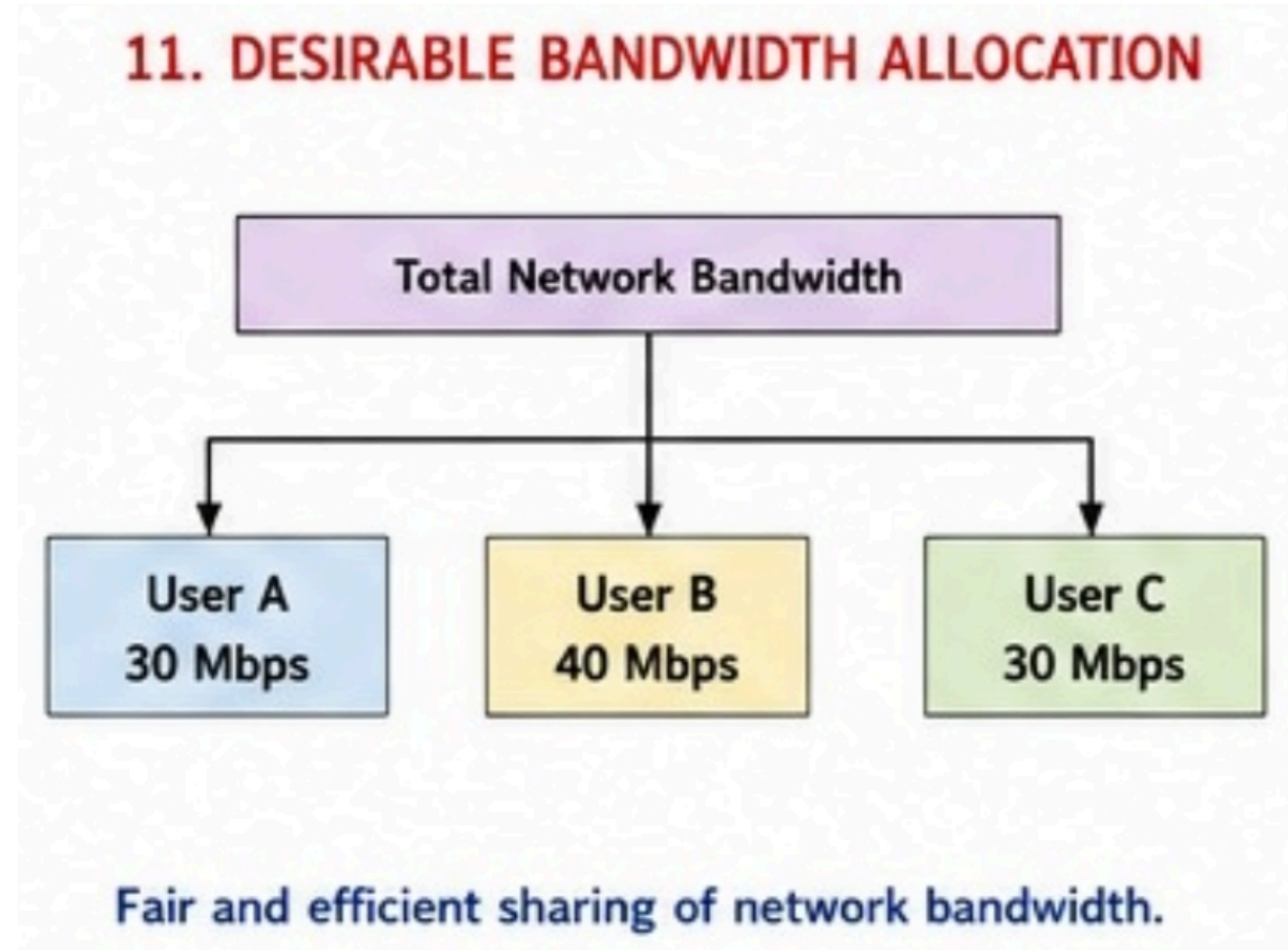


DESIRABLE BANDWIDTH ALLOCATION

- Efficient sharing of network bandwidth
- Fair allocation of available resources
- Prevents bandwidth starvation
- Improves network utilization
- Supports stable communication
- Important for congestion management
- Balances efficiency and fairness

Goals

- Efficiency
- Fairness
- Stability

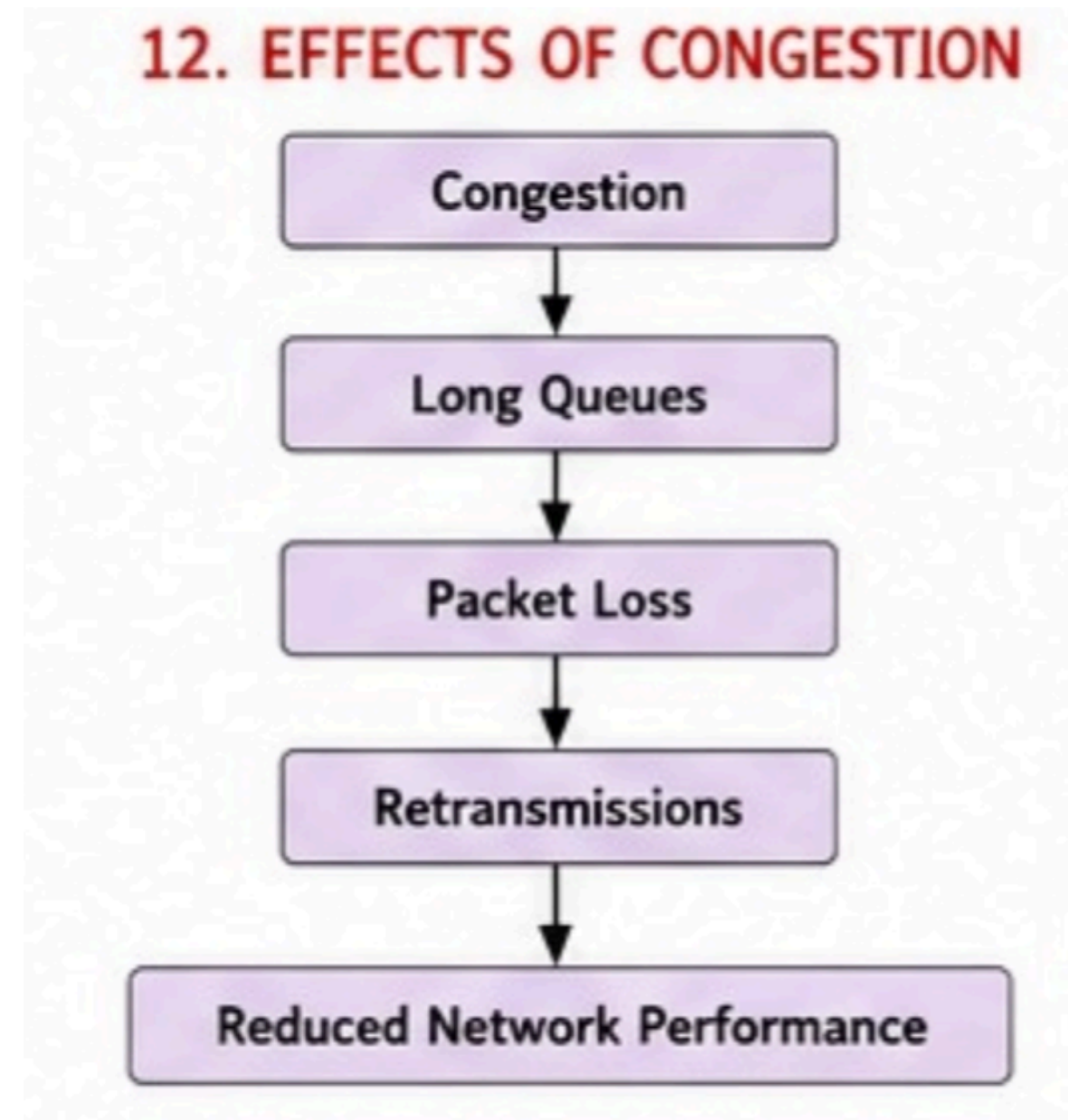


EFFECTS OF CONGESTION

- Transmission delay increases
- Packet loss occurs frequently
- Router queues become longer
- Retransmissions increase network traffic
- Throughput reduces significantly
- Communication becomes slower
- Overall network performance decreases

Important Effects

- Queue build-up
- Packet dropping
- Network slowdown



THANK YOU